

ScholarPath — Enhanced Entity–Relationship Diagram (EERD)

Data model report (Elmasri / Chen notation)

ScholarPath Graduation Project

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Abstract

This report documents the conceptual data model of the ScholarPath platform as an **Enhanced Entity–Relationship Diagram** using the **Chen / Elmasri** notation. The model was reverse-engineered from the live source code: the Entity Framework Core model snapshot (`ApplicationDbContextModelSnapshot.cs`), the domain entities under `ScholarPath.Domain/Entities`, and cross-checked against the five module SRS documents. The platform comprises **48 first-class domain entities** plus 7 ASP.NET Identity tables (`ApplicationUser`, `ApplicationRole` and the 5 standard junctions) = **55 EF-managed entities** (60 if the model snapshot’s standard Identity child tables are all counted); for legibility they are presented as one subject-area *context* view, a USER *specialization* (the “Enhanced” part), and six per-area EER diagrams. The full relational schema appears in the companion *Relational Mapping* report.

Contents

1	Notation legend	1
2	The USER specialization (the “Enhanced” part)	2
3	Subject-area context	3
4	Identity, Access & Profile	3
5	Scholarships, Applications & Documents	4
6	Consultant Booking, Payments & Ratings	5
7	Community & Chat	6
8	Resources Hub & Notifications	7
9	AI, Knowledge Base, Platform & Cross-cutting	8
A	Consolidated conceptual EERD	9

1 Notation legend

The diagrams use the Chen notation of Elmasri & Navathe:

- **Rectangle** — an entity type. (Every ScholarPath table has its own surrogate key, so all entities here are *strong*; the existence dependence of detail rows is shown with total participation, not weak entities.)

- **Diamond** — a relationship type connecting two entities.
- **Ellipse** — an attribute; an underlined attribute is (part of) the primary key. A *double* ellipse is a **multivalued** attribute and a *dashed* ellipse is a **derived** attribute.
- **Cardinality ratio** — the labels 1, N, M on the edges of a relationship diamond (1:1, 1:N, M:N).
- **Participation** — total participation (every instance must take part) is drawn with a double line, partial participation with a single line.

A convention specific to this system

ScholarPath deliberately distinguishes two kinds of references, and the model records both honestly:

- A **database-enforced foreign key** — a real FOREIGN KEY constraint with a declared delete rule (Cascade / Restrict / SetNull).
- An **application-enforced “loose” reference** — a Guid column that semantically points at another row but has *no* FK constraint; integrity is guaranteed by application code. High-volume, audit, log and analytics tables (notifications, votes, chat, audit log, AI interactions, payments) use loose references on purpose, so that deleting or anonymising a user never cascade-breaks history.

2 The USER specialization (the “Enhanced” part)

Conceptually a **USER** is a super-type specialised into four sub-types by the role(s) granted to it. This is the EER feature that makes the diagram *enhanced*:

- **Overlapping** specialization — a user may hold more than one role at once (a Student may be upgraded to Consultant and keep both).
- **Partial** specialization — a freshly registered user is **Unassigned** and belongs to no sub-type until role selection / onboarding completes.

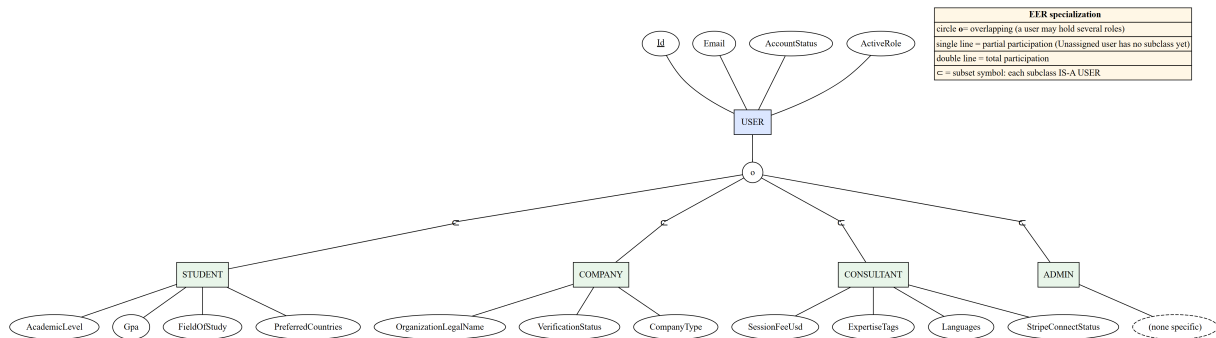


Figure 1: USER specialization in Elmasri EER notation — the circle marks an *overlapping* (o) specialization, the single line to USER marks *partial* participation, and each subset symbol ($\subset$) reads “IS-A”.

Sub-type	Role-specific attributes	Realised in
Student	AcademicLevel, FieldOfStudy, Gpa, PreferredCountries	UserProfiles
Company	OrganizationLegalName, VerificationStatus, CompanyAverageRating	UserProfiles
Consultant	SessionFeeUsd, ExpertiseTags, Languages, StripeConnectStatus	UserProfiles
Admin	(platform operator — no profile attributes)	UserRoles

Table 1: The four role sub-types of USER and where their attributes live.

Physical realisation. There is no table-per-subclass. The super-type is the **Users** table;

role membership is the **UserRoles** many-to-many join to **Roles** (seeded roles: **Admin**, **Student**, **Company**, **Consultant**, **Unassigned**). All role-specific attributes are columns on the single **UserProfiles** table (a single-table specialization), NULL when not applicable.

3 Subject-area context

Every subject area connects through the central **USERS** entity. Each box below is detailed in its own EER diagram in Sections 4–9. The dashed link marks the cross-area “paid review” flow (a Student pays to have a Company review their application).

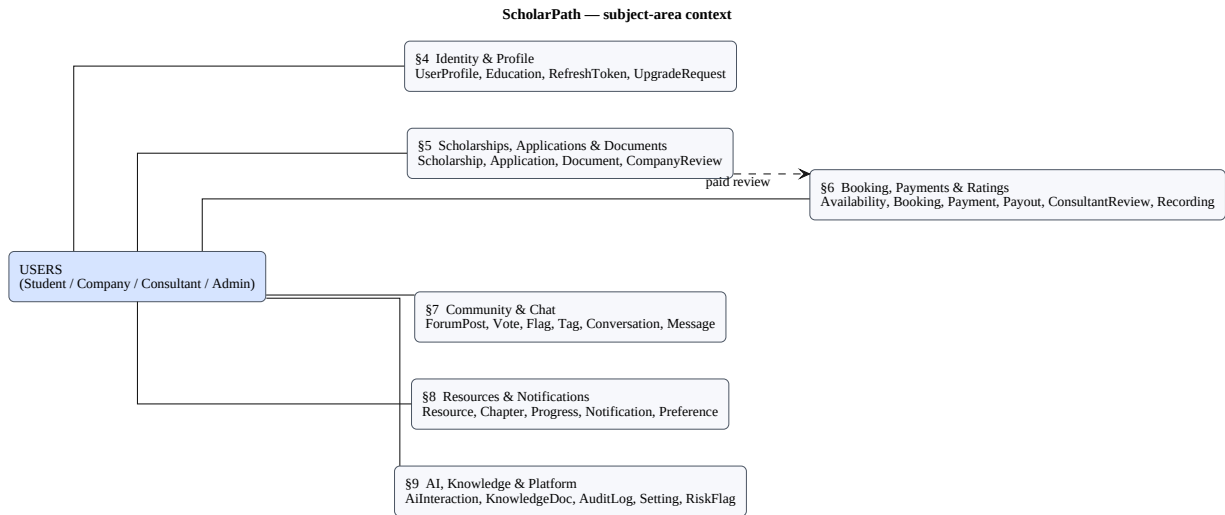


Figure 2: Subject-area context — everything radiates from USERS.

4 Identity, Access & Profile

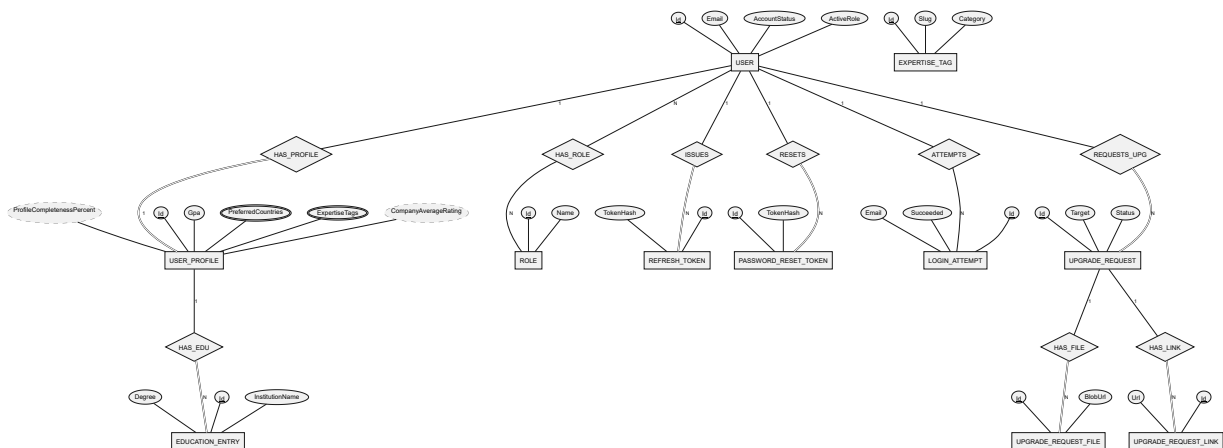


Figure 3: EERD — Identity, Access & Profile (Chen notation).

Key entities & relationships.

- **USER** 1:1 **USER_PROFILE** (profile total, user partial; FK, cascade).
- **USER** *M*:*N* **ROLE** through **UserRoles** (ASP.NET Identity).
- **USER_PROFILE** 1:*N* **EDUCATION_ENTRY** (total participation; cascade).
- **USER** 1:*N* **REFRESH_TOKEN**, **PASSWORD_RESET_TOKEN** (session / reset credentials, cascade).

- ## 5 Scholarships, Applications & Documents

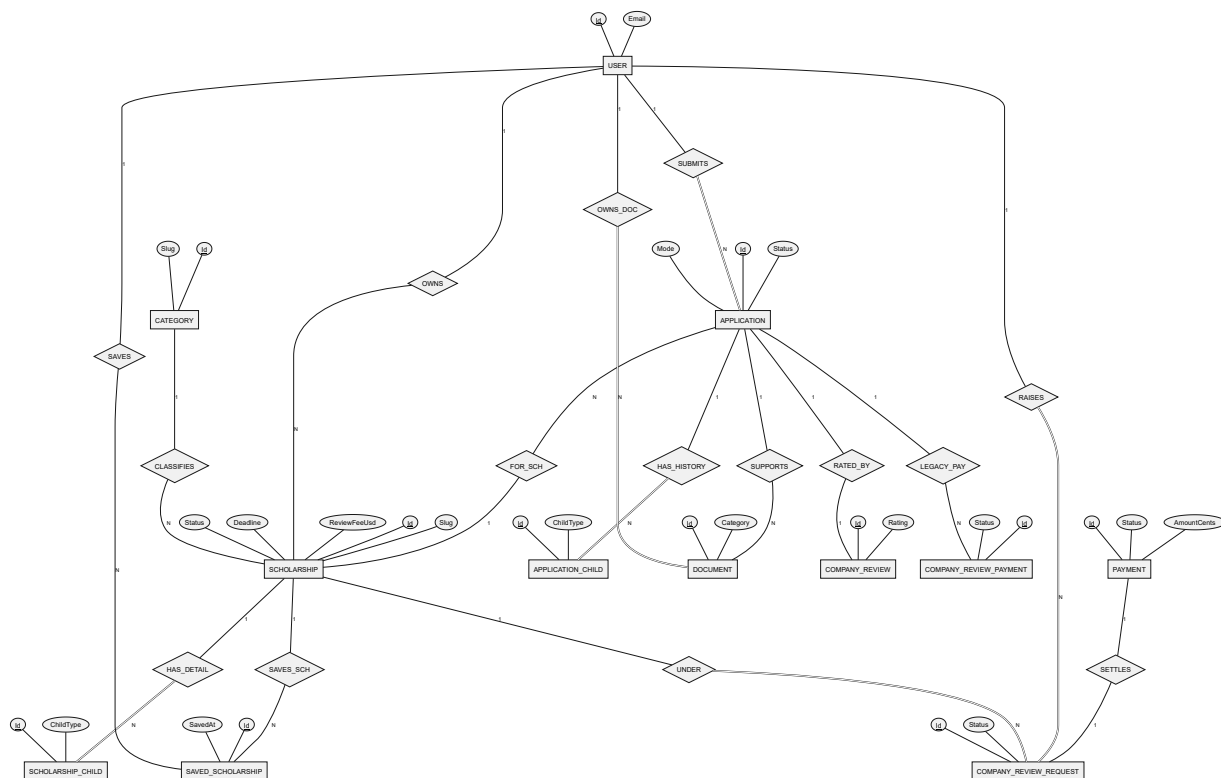


Figure 4: EERD — Scholarships, Applications & Documents.

- CATEGORY 1:*N* SCHOLARSHIP; USER (company) 1:*N* SCHOLARSHIP (owner may be null for admin / external listings).
- USER (student) 1:*N* APPLICATION *N*:1 SCHOLARSHIP (the scholarship link is optional — external trackers carry none).
- USER *M*:*N* SCHOLARSHIP via SAVED_SCHOLARSHIP (a bookmark; one per pair).
- APPLICATION 1:1 COMPANY_REVIEW (one review per finalised application); DOCUMENT optionally backs an application.
- COMPANY_REVIEW_REQUEST is the paid-support engagement linking Student, Company, Scholarship and a PAYMENT.

Business rule (FR-057). A filtered unique index enforces at most one *active* in-app application per student per scholarship.

6 Consultant Booking, Payments & Ratings

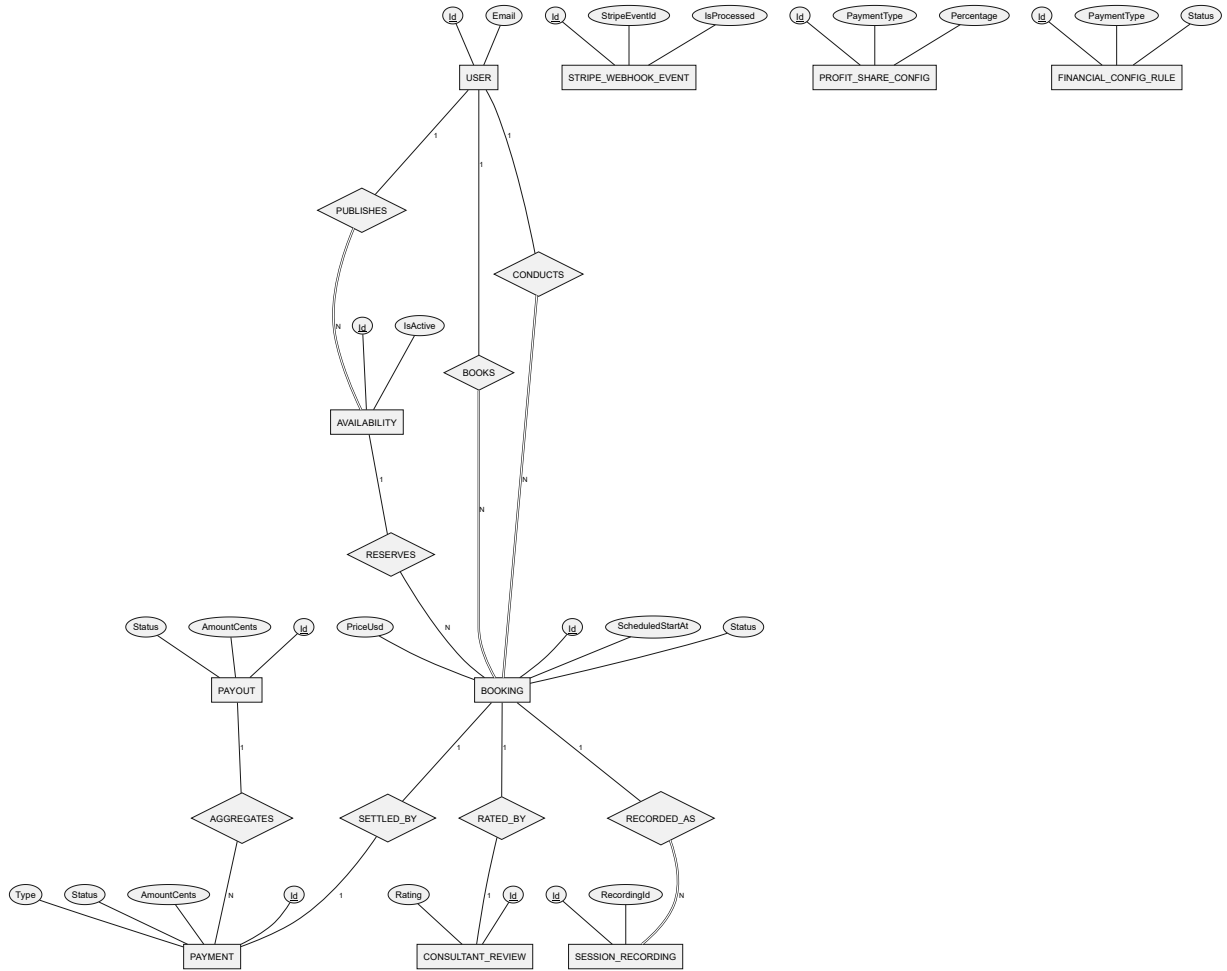


Figure 5: EERD — Booking, Payments, Ratings & Recording.

Key entities & relationships.

- **USER** (consultant) 1:*N* **AVAILABILITY**; a **BOOKING** is placed by a student and conducted by a consultant (two role FKs to **USER**).
- **BOOKING** 1:1 **PAYMENT** (settled-by) and 1:1 **CONSULTANT_REVIEW** (one review per completed booking).
- **BOOKING** 1:*N* **SESSION_RECORDING** (video recordings).
- **PAYOUT** aggregates many **PAYMENTS**. *Money is stored as integer cents*; **PAYMENT**/**PAYOUT** use loose references (no DB FK) to stay decoupled from the principals for audit retention.

7 Community & Chat

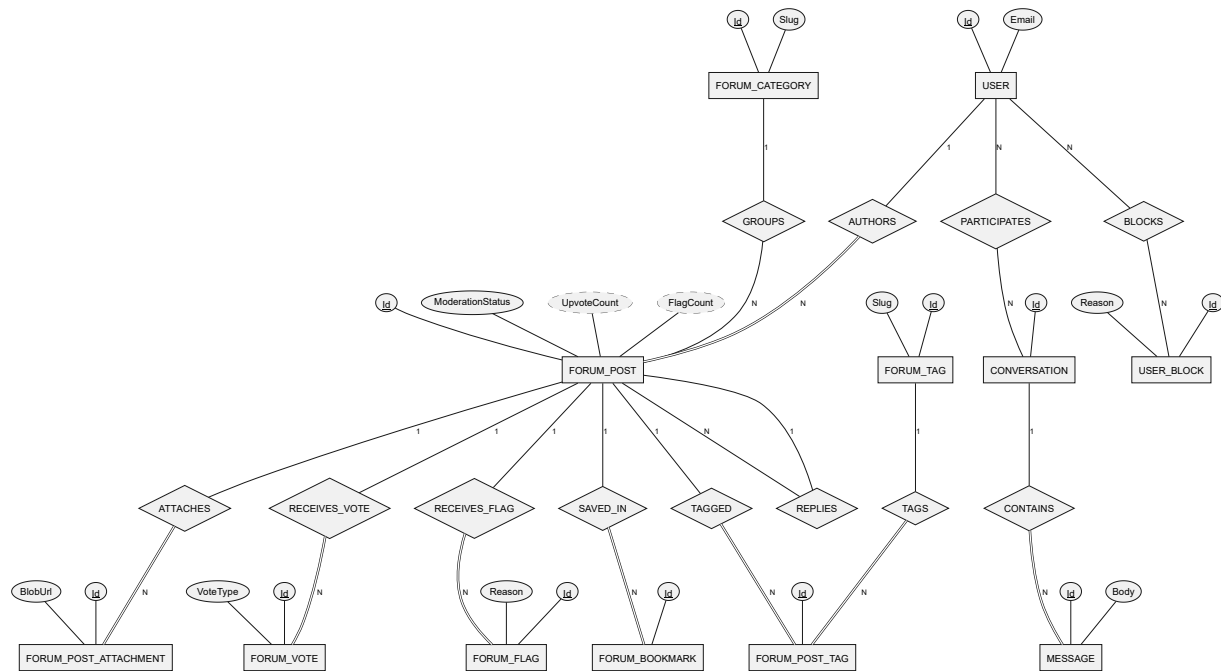


Figure 6: EERD — Community & Chat.

Key entities & relationships.

- **FORUM_POST** is self-referencing (threaded replies) and is grouped by **FORUM_CATEGORY**; it receives **FORUM_VOTE**, **FORUM_FLAG** and **FORUM_BOOKMARK** (each unique per user per post).
- **FORUM_POST** $M:N$ **FORUM_TAG** via the **FORUM_POST_TAG** junction.
- **CONVERSATION** is a one-to-one direct-message channel (unique participant pair) containing many **MESSAGES**.

8 Resources Hub & Notifications

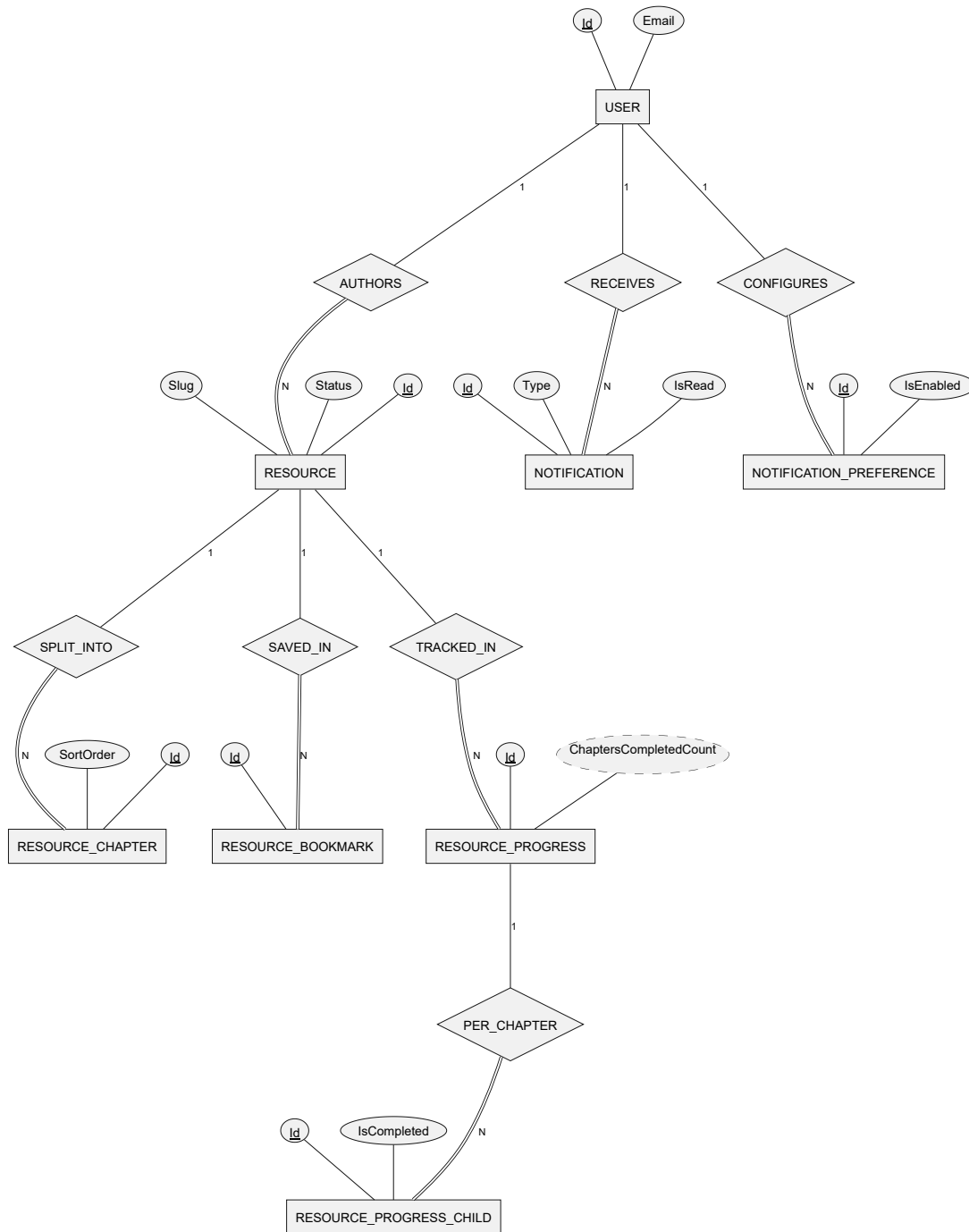


Figure 7: EERD — Resources Hub & Notifications.

Key entities & relationships.

- **RESOURCE** is split into **RESOURCE_CHAPTERs**; users bookmark resources and their per-chapter progress is tracked (**RESOURCE_PROGRESS** / **RESOURCE_PROGRESS_CHILD**).
- **USER** 1:N **NOTIFICATION** and **NOTIFICATION_PREFERENCE** (one toggle per user × type × channel); both via loose references.

9 AI, Knowledge Base, Platform & Cross-cutting

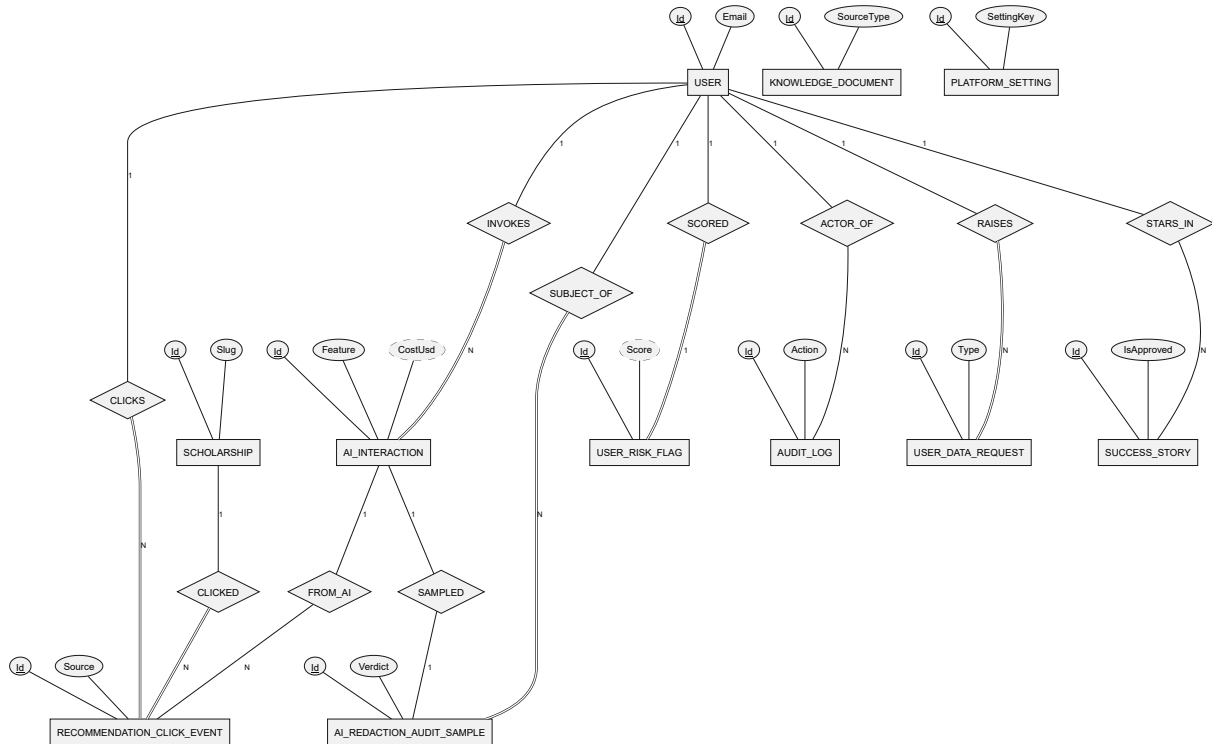


Figure 8: EERD — AI, Knowledge, Platform & Cross-cutting.

Key entities & relationships.

- **AI_INTERACTION** logs every AI call (cost + tokens); **RECOMMENDATION_CLICK_EVENT** ties a user, a scholarship and an optional interaction; **AI_REDACTION_AUDIT_SAMPLE** is a 1:1 compliance sample of an interaction.
- **USER_RISK_FLAG** is a 1:1 churn-risk row per user (written by a Power BI dataflow). **KNOWLEDGE_DOCUMENT** stores the RAG corpus with an in-row vector embedding and a *polymorphic* source reference.
- **AUDIT_LOG** uses a polymorphic (**TargetType**, **TargetId**) reference — a soft link to any entity, not a single FK.

For completeness, the principal entities of the whole platform on one canvas (dense by nature — the per-area diagrams above are the readable views).

